Eco 202 - Fall 2026

Class 11

- X, Y random variables

$X \in \{0, 1\} \Rightarrow X \sim \text{Bernoulli}(p)$, $p = \mathbb{P}(X = 1)$

- Expectations

$$\mathbb{E}[g(X)] = \sum_{x \in X} g(x) \cdot \mathbb{P}[X = x] \quad \text{(discrete)}$$

$$= \int g(x) \underbrace{f(x)}_{\text{density}} dx \quad \text{(continuous)}$$

- Density for a continuous random variable

$$f(x) = \frac{d}{dx} F(x)$$

probability distribution fn (pdf)

CDF

$$\underbrace{F(x)}_{\text{cumulative distribution fn.}} = \mathbb{P}[X \leq x]$$

- Properties for Expectation

$$\mathbb{E}[a \cdot X + b \cdot Y]$$

not random — random

$$\sum_{x,y} = \sum_{x} \sum_{y} = \sum_{y} \sum_{x}$$

$$= \sum_{x,y} (a \cdot x + b \cdot y) \cdot \mathbb{P}[X = x, Y = y]$$

$$= \sum_{x,y} a \cdot x \cdot \mathbb{P}[X = x, Y = y] + \sum_{x,y} b \cdot y \cdot \mathbb{P}[X = x, Y = y]$$

(hint: $\sum_{x,y} g(x,y) = \sum_{x} \sum_{y} g(x,y) = \sum_{y} \sum_{x} g(x,y)$)

$$= \sum_{x} a \cdot x \cdot \underbrace{\sum_{y} \mathbb{P}[X = x, Y = y]}_{\mathbb{P}[X = x]} + \sum_{y} b \cdot y \cdot \underbrace{\sum_{x} \mathbb{P}[X = x, Y = y]}_{\mathbb{P}[Y = y]}$$

Then :

$$\mathbb{E}[aX + bY] = \mathbb{E}[aX] + \mathbb{E}[bY]$$

$$= a \cdot \mathbb{E}[X] + b \cdot \mathbb{E}[Y]$$

Property for Variance

$(a+b)^2 = a^2 + 2ab + b^2$

$$\mathbb{V}[a \cdot X + b \cdot Y]$$

$$= \sum_{x,y} \left(ax + by - \mathbb{E}[a \cdot X + b \cdot Y] \right)^2 \cdot \mathbb{P}[X = x, Y = y]$$

$\vdots$ (You try! Seriously, try!)

$$= \mathbb{V}[aX] + 2\,\mathrm{Cov}(aX, bY) + \mathbb{V}[bY]$$

Then:

$$V[aX + bY]$$

$$= a^2 V[X] + b^2 V[Y] + 2ab \cdot Cov(X, Y)$$

If X and Y are independent, then $Cov(X, Y) = 0$

Then:

$$V[aX + bY] = a^2 V[X] + b^2 V[Y]$$

Sampling distribution

X_1 = wage for unit 1

X_2 = wage for unit 2

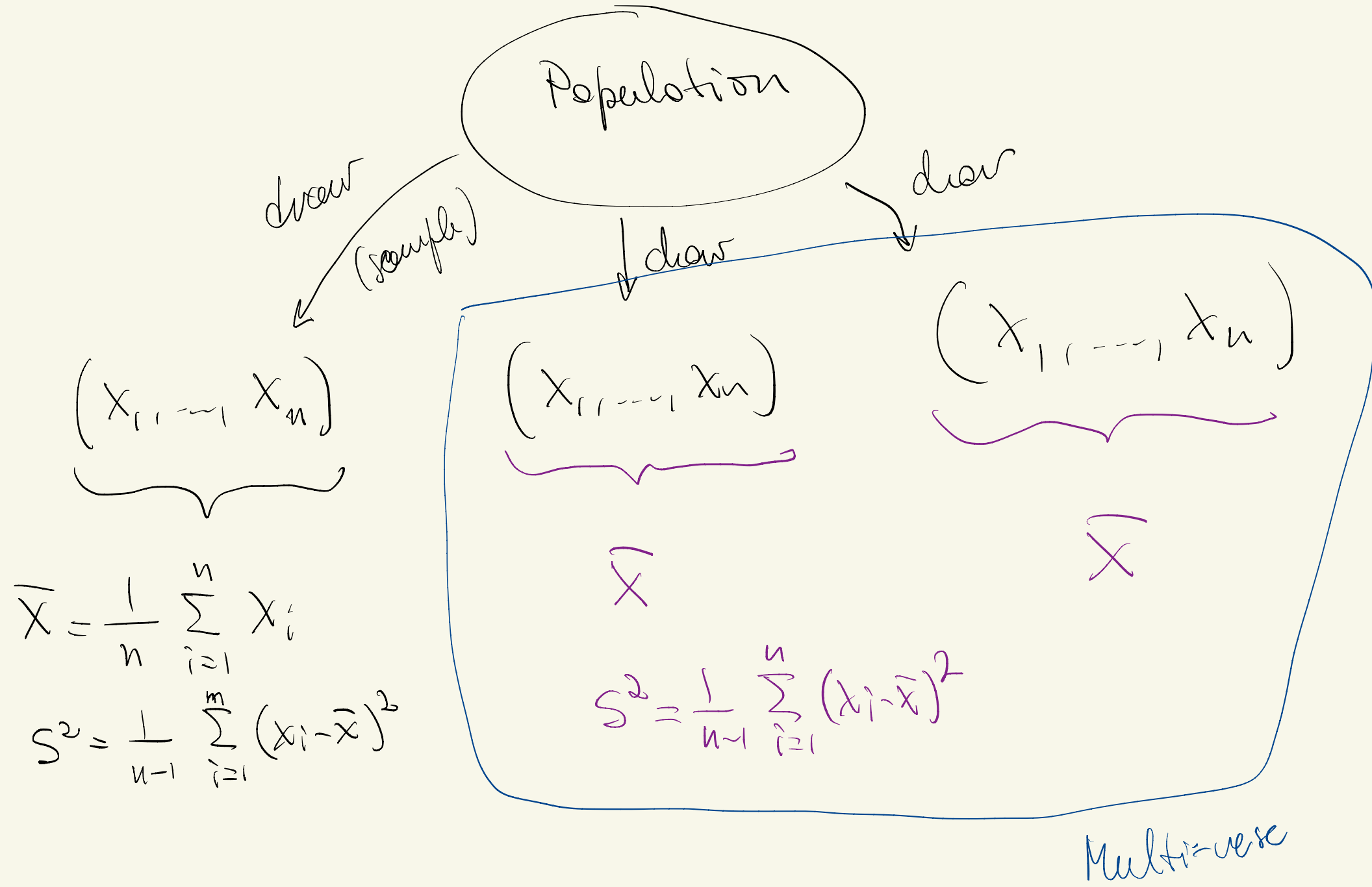
$\vdots$

X_n = wage for unit n

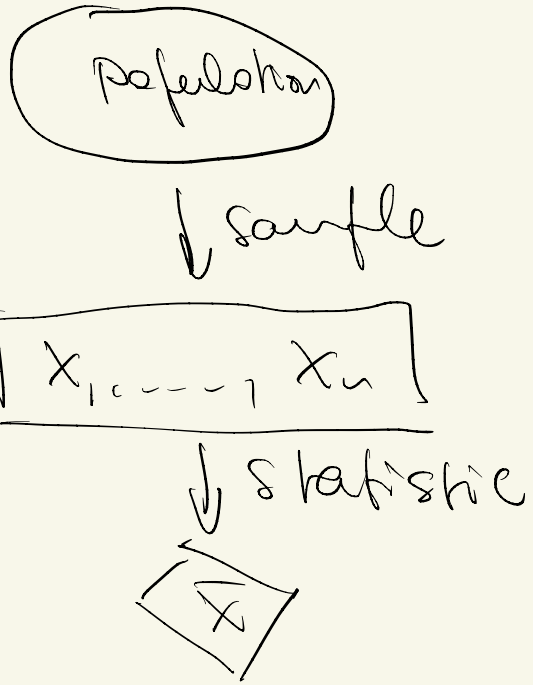
A dataset contains the realizations of these random variables: $(X_1, \dots, X_n)$

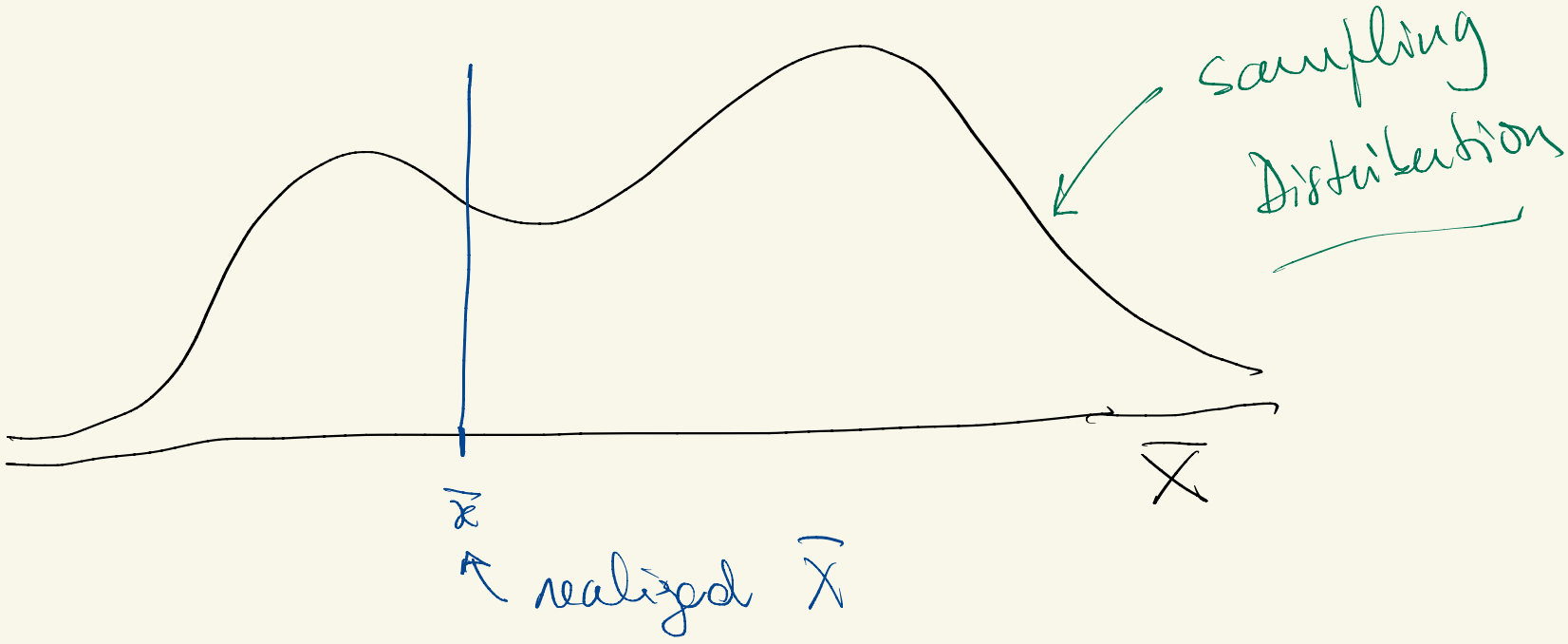
thus: $X_1 = x_1$, $X_2 = x_2$, ..., $X_n = x_n$

this is the data that you have!

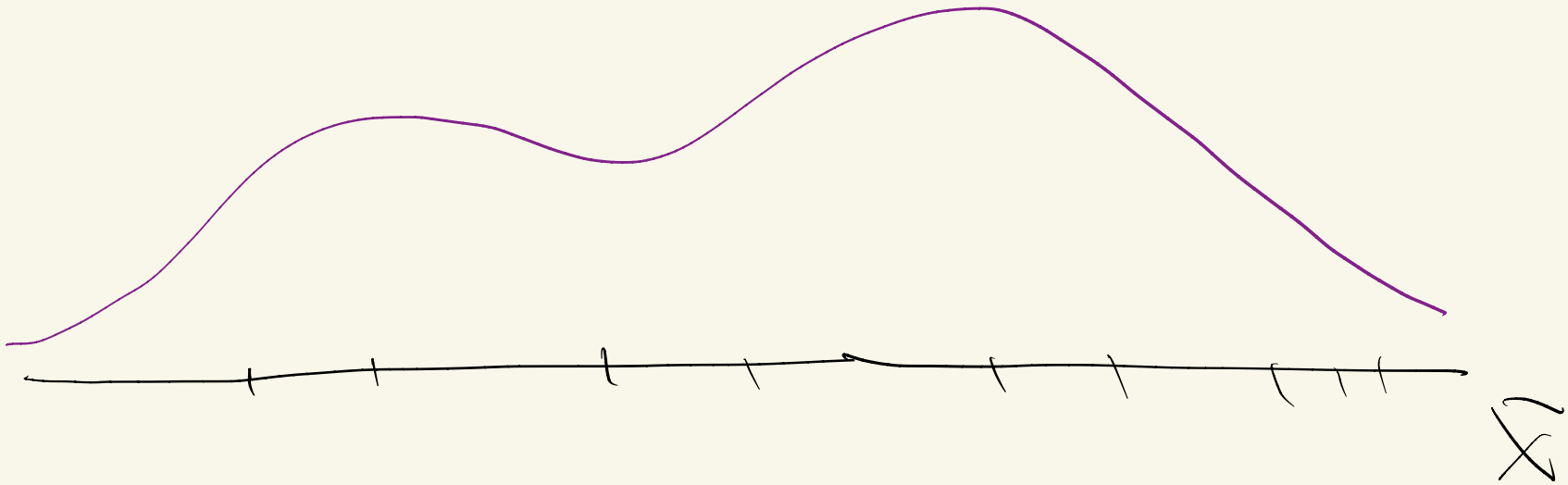

Population
draw
(sample)
draw
draw
draw
$(x_1, \ldots, x_n)$
$(x_1, \ldots, x_n)$
$(x_1, \ldots, x_n)$
$\bar{x}$
$\bar{x}$
$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$
$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$
$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2$
Multi-verse

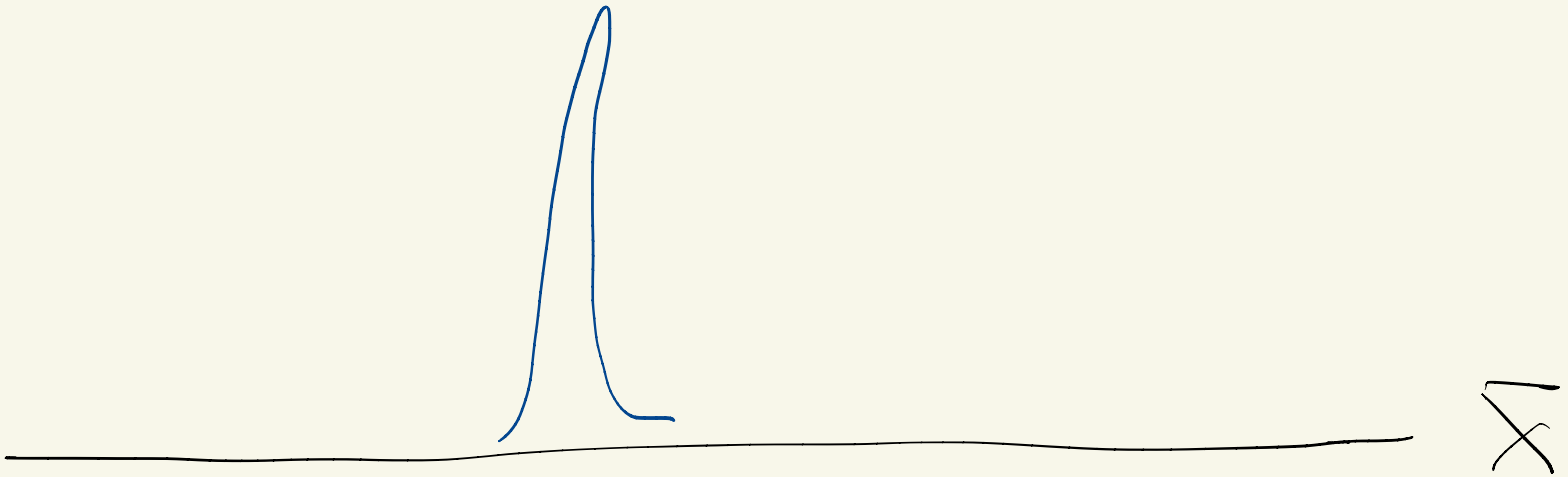
- Population
- Sampling
- Statistic
- Sampling Distribution





- two examples of sampling distributions





- Setup: $X_1, X_2, \ldots, X_n$ random variables
- Assume: independent, identically distributed } i.i.d.
- this implies: $\mathbb{E}[X_i] = \mu$ $\quad \mathbb{V}[X_i] = \sigma^2$
- look at sample average

$$\bar{X} = \frac{1}{n} \sum_{i=1}^{n} X_i$$

$\bar{X}$ is random $\Rightarrow$ it has a sampling distribution

(1) Expected value

$$\mathbb{E}[\bar{X}] = \mathbb{E}\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$$

$$= \frac{1}{n}\,\mathbb{E}\left[\sum_{i=1}^{n} X_i\right] \qquad \left(\frac{1}{n} \text{ is a constant}\right)$$

$$= \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[X_i] \qquad \left(\mathbb{E}[X+Y] = \mathbb{E}[X]+\mathbb{E}[Y]\right)$$

$$= \frac{1}{n}\sum_{i=1}^{n}\mu = \mu$$

(2) Variance

$$V[\bar{X}] = V\left[\frac{1}{n}\sum_{i=1}^{n} X_i\right]$$

$$= \frac{1}{n^2} V\left[\sum_{i=1}^{n} X_i\right] \qquad \left(\frac{1}{n} \text{ is a constant}\right)$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} V[X_i] \qquad (\text{independence})$$

$$= \frac{1}{n^2} \sum_{i=1}^{n} \sigma^2 = \frac{\sigma^2}{n}$$